

sol1_enhancer_opencode

Same loop. Jail is **permission deny**, not a child process.

Native `.opencode/` discovery. Not a copy of `.claude/`. Not an OpenCode JS plugin. No `plugin.json`.

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What you will build

PIECE	PATH
Judge	<code>.opencode/agents/enhancer-judge.md</code>
Doer	<code>.opencode/agents/enhancer-doer.md</code>
Skill	<code>.opencode/skills/enhancer-loop/SKILL.md</code>
Scripts	<code>.opencode/skills/enhancer-loop/scripts/check_{fields,stop}.py</code>
Command	<code>.opencode/command/enhancer-loop.md</code>
Permissions	<code>opencode.json</code>

Saturday FALL-BEHIND.md still says "No OpenCode answer exists yet." That paragraph is stale. This folder is the answer.

Why OpenCode is a different lesson

Claude omits Write from a YAML tool list. Codex starts a second process. OpenCode puts `permission: edit: deny` on the subagent.

`--auto` does not override explicit deny. That is the claim to prove.

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Learning objectives

- Configure OpenCode agents with `mode: subagent`
- Deny edit, bash, and task on the judge
- Invoke headless via `--command enhancer-loop`
- Probe the jail with `cksum` before a live poll
- Point Actions at `ENHANCER_BACKEND=opencode`

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Starting architecture

```
opencode run --auto --command enhancer-loop
└─ agent: build (orchestrator, default)
    Task tool
        └─ enhancer-judge    permission.edit: deny
        └─ enhancer-doer     permission.edit: deny
```

--agent enhancer-judge fails: "is a subagent, not a primary agent." That is correct. The orchestrator is build.

Judge frontmatter. This is the type system

```
---  
description: Grade one enhancer ticket. Return JSON only.  
mode: subagent  
permission:  
  edit: deny  
  bash: deny  
  task: deny  
---
```

The body then says: if asked to write or edit, refuse and still return the JSON grade of the file as it currently is.

`opencode.json` and the command file

```
{ "permission": { "external_directory": "allow" } }
```

Without that, writes to `../../work/northwind-field-crm` become asks. Headless `--auto` would then stall.

`.opencode/command/enhancer-loop.md` pins `agent: build` and substitutes `$ARGUMENTS`.

Headless argv

```
opencode run
  --dir {{.TASKFILE_DIR}}
  --auto
  --command enhancer-loop
  --
  --repo {{.TARGET}} {{.CLI_ARGS}}
  < /dev/null
```

No built-in loop skill. No self-reinvoke. First poll is about six minutes. Cap it:

```
timeout 360 task run --
```


Jail probe before a live poll

Spawn the judge. Ask it to edit a file. `cksum` of that file must be identical.

If the file changed, `--auto` overrode deny and the port is wrong. Stop. Do not run `task create-test-tickets` yet.

Commands

```
cd solutions/sol1_enhancer_opencode  
cp config.json.example config.json  
task clone  
python3 .opencode/skills/enhancer-loop/scripts/check_fields.py --demo  
python3 .opencode/skills/enhancer-loop/scripts/check_stop.py --demo  
task create-test-tickets  
timeout 360 task run --
```

Expected result

Same as Saturday: stubs become real fields, `enhanced` after a rewrite, marked comment, state file.

`check_fields.py` still computes ready. The permission block did not have to change the rubric.

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Troubleshooting

SYMPTOM	CAUSE	FIX
"is a subagent, not a primary"	<code>--agent enhancer-judge</code>	orchestrator is <code>build</code>
Ask-loop on CRM writes	missing <code>external_directory</code>	set <code>allow</code> in <code>opencode.json</code>
Judge edited the ticket	deny not applied	probe with <code>cksum</code>
Six-minute hang	first poll is slow	<code>timeout 360</code>
FALL-BEHIND says no answer	stale doc	this folder is the answer

Validation

- [] Judge YAML has `edit: deny`
- [] Jail probe: file checksum unchanged
- [] `--demo` scripts pass
- [] One poll labels `enhanced` after a rewrite
- [] Marker on the comment

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GitHub Actions

`ENHANCER_BACKEND=opencode`. The job needs `opencode` on the runner.

Copy `labs/lab1_enhancer/workflows/enhance-on-issue.yml` onto **your** CRM fork. Do not enable it on the instructor repo.

Skip comments that contain `<!-- enhancer-loop -->`.

Recap

OpenCode isolation is a permission block on the subagent.

Same rubric. Same marker. Same exits. The loop did not have to change to make it run.

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